

Solution of Exercise 1

(a) We can rewrite A as $A_1 \cup A_2$, where

$$A_1 = \left\{ \frac{1}{2} + \frac{n}{2n+1} : n \in \mathbb{N}_0 \right\} \quad \text{and} \quad A_2 = \left\{ 2 - \frac{1}{n+2} : n \in \mathbb{N}_0 \right\}.$$

We proved (exercise 11 set 2) that A_1 is bounded since

$$\forall n \in \mathbb{N}_0 : \frac{1}{2} \leq \frac{1}{2} + \frac{n}{2n+1} \leq 1,$$

$$\forall x = \frac{1}{2} + \frac{n}{2n+1} \in A_1, \quad \frac{1}{2} \leq x < 1.$$

with $\sup A_1 = 1$ and $\inf A_1 = \frac{1}{2}$.

Let us focus now on A_2 . The set A_2 is bounded, because

$$\forall n \in \mathbb{N}_0 : \frac{3}{2} \leq 2 - \frac{1}{n+2} < 2. \quad (1)$$

(b) We prove that $\sup A_2 = 2$ and $\inf A_2 = \frac{3}{2}$. We proceed with $\sup A_2$.

(i) From (1), we have $\forall x \in A_2 : x < 2$.

(ii) We show that $\forall \varepsilon > 0, \exists x_\varepsilon \in A : x_\varepsilon > 2 - \varepsilon$. Indeed,

$$\begin{aligned} x > 2 - \varepsilon &\Leftrightarrow 2 - \frac{1}{n+2} > 2 - \varepsilon \\ &\Leftrightarrow \varepsilon > \frac{1}{n+2} \\ &\Leftrightarrow n > \frac{1}{\varepsilon} - 2. \end{aligned}$$

That is, for any $\varepsilon > 0$ we can choose $n_\varepsilon \in \mathbb{N}_0$ satisfying $n_\varepsilon > \frac{1}{\varepsilon} - 2$ for which

$$A_2 \ni x_\varepsilon = 2 - \frac{1}{n_\varepsilon + 2} > 2 - \varepsilon.$$

Thus $\sup A_2 = 2$.

Next we proceed with $\inf A_2$. We have

(i) From (1), we have $\forall x \in A_2 : x \geq \frac{3}{2}$.

(ii) We show that $\forall \varepsilon > 0, \exists x_\varepsilon \in A_2 : x_\varepsilon < \frac{3}{2} + \varepsilon$. Indeed, for $0 < \varepsilon < 1/2$, we have

$$\begin{aligned} x < \frac{3}{2} + \varepsilon &\Leftrightarrow 2 - \frac{1}{n+2} < \frac{3}{2} + \varepsilon \\ &\Leftrightarrow n+2 < \frac{1}{0.5 - \varepsilon} \\ &\Leftrightarrow n < \frac{1}{0.5 - \varepsilon} - 2. \end{aligned}$$

That is, for any $0 < \varepsilon < 1/2$ (the case $\varepsilon \geq 1/2$ is trivial), we can choose $n_\varepsilon \in \mathbb{N}_0$ satisfying

$$n_\varepsilon < \frac{1}{0.5 - \varepsilon} - 2$$

for which

$$A_2 \ni x_\varepsilon = 2 - \frac{1}{n_\varepsilon + 2} < \frac{3}{2} + \varepsilon.$$

Thus $\inf A_2 = \frac{3}{2}$.

We know that

$$\sup(A_1 \cup A_2) = \max(\sup A_1, \sup A_2) \quad \text{and} \quad \inf(A_1 \cup A_2) = \min(\inf A_1, \inf A_2).$$

Therefore

$$\sup(A_1 \cup A_2) = 2 \quad \text{and} \quad \inf(A_1 \cup A_2) = \frac{1}{2}.$$

(c) For $n = 0$, we have

$$A \ni x_0 = \frac{1}{2} + \frac{0}{2(0) + 1} = \frac{1}{2} = \inf A.$$

Hence $\min A = \inf A$.

Now, assume that $\exists n \in \mathbb{N}_0$ such that

$$2 - \frac{1}{n+2} = \sup A = 2.$$

Then, $1/(n+2) = 0$, which is a contradiction. Thus, $\max A$ does not exist.

Solution of Exercise 2

First Method. We have

$$\overline{\left(\frac{x + u_i}{1 + xu_i}\right)} = \frac{x + \overline{u_i}}{1 + x\overline{u_i}} = \frac{x + \frac{1}{u_i}}{1 + \frac{x}{u_i}} = \frac{1 + xu_i}{x + u_i}, \quad i = 1, \dots, n.$$

Hence

$$\overline{Z} = \sum_{i=1}^n \overline{\left(\frac{x + u_i}{1 + xu_i}\right)} \sum_{i=1}^n \overline{\left(\frac{1 + xu_i}{x + u_i}\right)} = \left(\sum_{i=1}^n \frac{1 + xu_i}{x + u_i}\right) \left(\sum_{i=1}^n \frac{x + u_i}{1 + xu_i}\right) = Z.$$

Then Z is a real number.

Second Method. We have

$$Z = \left(\sum_{i=1}^n \frac{1+xu_i}{x+u_i} \right) \left(\sum_{i=1}^n \frac{x+u_i}{1+xu_i} \right) = \left(\sum_{i=1}^n \frac{1+xu_i}{x+u_i} \right) \left(\overline{\sum_{i=1}^n \frac{1+xu_i}{x+u_i}} \right) = \left| \sum_{i=1}^n \frac{1+xu_i}{x+u_i} \right|^2$$

then Z is a real number.

Solution of Exercise 3

First Method. Remark that

$$\begin{aligned} (x_1 + 2x_2 + \dots + nx_n) &= (x_1 + x_2 + \dots + x_n) \\ &\quad + (x_2 + x_3 + \dots + x_n) + (x_3 + \dots + x_n) + \dots + (x_{n-1} + x_n) + x_n. \end{aligned}$$

Since $x_1 + \dots + x_n = 0$, we have $x_2 + \dots + x_n = -x_1 \dots$

First case. If $n = 2p$, then $\frac{n^2}{4} = p^2$ and so, $E\left(\frac{n^2}{4}\right) = p^2 = \frac{n^2}{4}$. In this case we have

$$\begin{aligned} |x_1 + 2x_2 + \dots + nx_n| &\leq |x_1 + x_2 + \dots + x_{2p}| + |x_2 + \dots + x_{2p}| + \dots + |x_p + \dots + x_{2p}| \\ &\quad + |x_{p+1} + \dots + x_{2p}| \dots + |x_{2p-1} + x_{2p}| + |x_{2p}| \\ &= 0 + |-x_1| + |-x_1 - x_2| + \dots + |-x_1 + \dots - x_{p-1}| \\ &\quad + |x_{p+1} + \dots + x_{2p}| \dots + |x_{2p-1} + x_{2p}| + |x_{2p}| \\ &\leq 0 + 1 + 2 + \dots + (p-1) + p + (p-1) + \dots + 1 \\ &= 2\frac{p(p-1)}{2} + p = p^2 = E\left(\frac{n^2}{4}\right). \end{aligned}$$

Second case. If $n = 2p + 1$, then $\frac{n^2}{4} = p^2 + p + \frac{1}{4}$ and so

$$E\left(\frac{n^2}{4}\right) = p^2 + p = \frac{n^2 - 1}{4}.$$

In this case we have

$$\begin{aligned} |x_1 + 2x_2 + \dots + nx_n| &\leq |x_1 + x_2 + \dots + x_{2p+1}| + \dots + |x_{p+1} + \dots + x_{2p+1}| \\ &\quad + |x_{p+2} + \dots + x_{2p+1}| \dots + |x_{2p+1}| \\ &= 0 + |-x_1| + |-x_1 - x_2| + \dots + |-x_1 + \dots - x_p| \\ &\quad + |x_{p+2} + \dots + x_{2p+1}| \dots + |x_{2p+1}| \\ &\leq 0 + 1 + 2 + \dots + (p-1) + p + p + (p-1) + \dots + 1 \\ &= 2\frac{p(p+1)}{2} = p^2 + p = E\left(\frac{n^2}{4}\right). \end{aligned}$$

The above shows that

$$\forall n \in \mathbb{N} : |x_1 + 2x_2 + \dots + nx_n| \leq E\left(\frac{n^2}{4}\right)$$

Second Method

First, we have

$$\left\lceil \frac{n^2}{4} \right\rceil = \begin{cases} p^2, & \text{if } n = 2p \\ p^2 + p, & \text{if } n = 2p + 1 \end{cases}$$

– If $n = 2p$

$$\begin{aligned} \left| \sum_{k=1}^n kx_k \right| &= \left| \sum_{k=1}^{2p} kx_k - p \sum_{k=1}^{2p} x_k \right| = \left| \sum_{k=1}^{2p} (k-p) x_k \right| \leq \sum_{k=1}^{2p} |(k-p) x_k| \\ &\leq \sum_{k=1}^{2p} |(k-p)| = \sum_{k=1}^{p-1} |(k-p)| + \sum_{k=p+1}^{2p} |(k-p)| \\ &= 2 \sum_{k=1}^{p-1} k = (p^2 - p) \leq \left\lceil \frac{n^2}{4} \right\rceil \end{aligned}$$

– If $n = 2p + 1$,

$$\begin{aligned} \left| \sum_{k=1}^n kx_k \right| &= \left| \sum_{k=1}^{2p+1} kx_k - (p+1) \sum_{k=1}^{2p+1} x_k \right| = \left| \sum_{k=1}^{2p+1} (k-p-1) x_k \right| \\ &\leq \sum_{k=1}^{2p+1} |(k-p-1) x_k| \leq \sum_{k=1}^{2p+1} |(k-p-1)| \\ &= \sum_{k=1}^p |k-p-1| + \sum_{k=p+2}^{2p+1} |k-p-1| \\ &= 2 \sum_{k=1}^p k = (p^2 + p) \leq \left\lceil \frac{n^2}{4} \right\rceil. \end{aligned}$$

Then for all $n \in \mathbb{N}$, we have

$$\left| \sum_{k=1}^n kx_k \right| \leq \left\lceil \frac{n^2}{4} \right\rceil.$$